

HUMED LIFESCIENCES

"Transforming patient access to Dental Implant treatments to those that need it the most"

JULY 2025



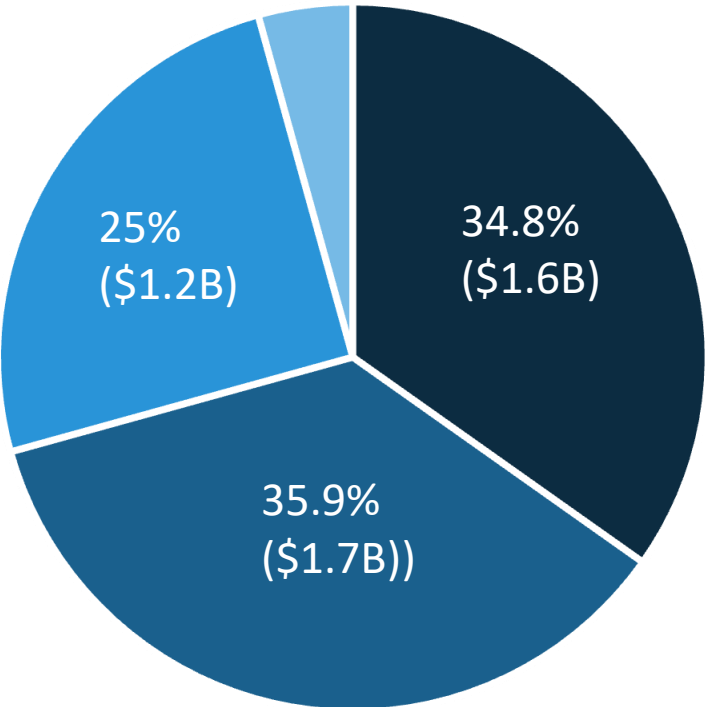
Fundamental Implant Breakthrough in 30+ Years



- Dental implants: Artificial tooth roots replacing missing teeth
- 1980s: Osseointegration breakthrough enabled titanium fusion
- Three decades of only minor surface modifications since then
- **HuMed delivers long-awaited structural advancement**

Thriving Market, Billion-Dollar Prizes

\$4.7B Global Dental Implant Market
8.0% CAGR annually and \$7.0B+ expected by 2030



■ North America ■ Europe ■ Asia-Pacific ■ Rest of World

Proven Success Stories
Industry giants with \$2-25B market capitalizations

Company	Revenue	Market Cap	Note
Straumann	~\$1.7B	\$18.5B	Global NO.1
Envista	~\$1.0B	\$3.1B	40% Implant
Ostem	~\$0.6B	\$2.3B	Asia NO.1
Dentsply Sirona	~\$0.5B	\$3.1B	Equipment Focus
Zimmer Biomet	~\$0.3B	\$25B	Orthopedics Focus

** Revenue = Estimated implant segment only | Market Cap = July 2025 data*

World's First Hybrid Dental Implant Technology

First-ever Injectable Implant for Bone Ingrowth Technology
US Patented Game-Changing Dental Implant

Advanced Injectable Implant System
*PCT for Global Patent Strategy

US patent application

Publication No. : US2012/0141955A1
Application No. : 13/310,934
Inventor : Do-Gyoon Kim
Assignee : The Ohio State University
Filed : 12/05/2011
Tech. Center/Art Unit : 3732

Attorney File No. : OSU 11001
Customer No. : 002555
Confirmation Number : 4075
For : Hybrid Dental Implant

Continuation-in-part from 13/310,934

Publication No. : US2015/0230889A1
Application No. : 14/704,295
Inventor : Do-Gyoon Kim
Assignee : The Ohio State University
Filed : 5/5/2015
Tech. Center/Art Unit : 3732

Attorney File No. : OSU 11001
For CIP of : Hybrid Dental Implant

US009757213B2

(12) **United States Patent**
Kim

(10) **Patent No.:** US 9,757,213 B2
(45) **Date of Patent:** Sep. 12, 2017

(54) **HYBRID DENTAL IMPLANT** 606/156, 157, 158, 304; 604/506, 174, 604/20, 501

(71) **Applicant:** Ohio State Innovation Foundation, Columbus, OH (US) See application file for complete search history.

(72) **Inventor:** Do-Gyoon Kim, Dublin, OH (US) **References Cited**

(73) **Assignee:** Ohio State Innovation Foundation, Columbus, OH (US) U.S. PATENT DOCUMENTS

(*) **Notice:** Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days. 5,192,282 A 3/1993 Danner A61C 8/0018 5,360,374 A 1/1994 Vlassis 433/165 6,048,204 A 4/2000 Khalife et al. 6,755,835 B2* 6/2004 Schulheim A61B 17/3685 606/304 (Continued)

(21) **Appl. No.:** 14/704,295 **OTHER PUBLICATIONS**

(22) **Filed:** May 5, 2015 Illmes, R.J., et al.; A 10-Year Prospective Study ofITI Dental Implants Placed in the Posterior Region. I: Clinical and Radiographic Results; Clin. Oral Implants Res., Dec. 2007, vol. 18(6); pp. 699-706. (Continued)

(65) **Prior Publication Data** US 2015/0230889 A1 Aug. 20, 2015

Related U.S. Application Data

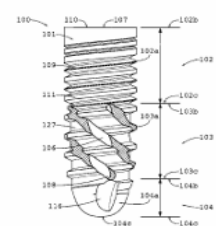
(63) Continuation-in-part of application No. 13/310,934, filed on Dec. 5, 2011, now abandoned. **Primary Examiner**—Cris L. Rodriguez **Assistant Examiner**—Mirayda A. Aponle

(66) Provisional application No. 61/419,333, filed on Dec. 3, 2010. (74) **Attorney, Agent, or Firm**—Meunier Carlin & Curfman LLC

(51) **Int. CL** A61C 8/00 (2006.01) **ABSTRACT**

(52) **U.S. CL** CPC: A61C 8/0022 (2013.01); A61C 8/0037 (2013.01) The present invention provides for a hybrid dental implant having a screw body with at least one external thread, which includes an upper portion with an open ceiling configured to receive one or more bone inducing agents and/or therapeutic agents, a middle portion having a hollow inner channel connected to side openings, and a lower portion with a closed floor. The bone inducing agents and/or therapeutic agents are introduced through the open ceiling into the hollow inner channel into the side openings, which enables the bone tissue ingrowth into the hollow inner channel and bone tissue outgrowth from the hollow inner channel to the surrounding bone. It also provides for a method of installing and using the hybrid dental implant thereof.

(58) **Field of Classification Search** CPC: A61C 8/0018; A61C 8/0024; A61C 8/0022; A61C 8/0037; A61C 8/009; F16B 2/065; F16B 5/0072; F16B 5/0004; F16B 25/00; F16B 19/08; F16B 2019/1009; F16B 2019/1018; F16B 7/18 USPC: 433/172-176, 201/1, 39, 135, 137, 138, 433/148, 149, 150, 153, 155, 156, 162; 15 Claims, 13 Drawing Sheets



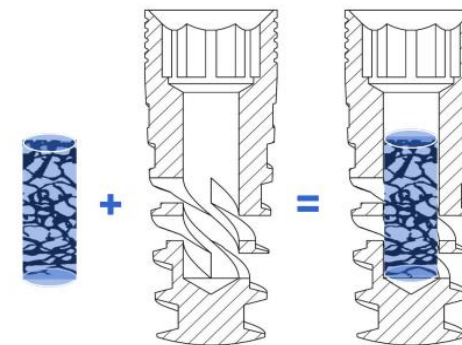
Patent PCT Submission

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Sheet No.

Box No. VIII (iv) DECLARATION: INVENTORSHIP (only for the purposes of the designation of the United States of America)
The declaration must conform to the following standardized wording provided for in Section 214: see Notes to Boxes Nos. VIII, VIII (i) to (iv) (in general) and the specific Notes to Box No. VIII (iv). If this Box is not used, this sheet should not be included in the request.

Declaration of inventorship (Rules 4.17(iv) and 51b(1)(a)(iv)) for the purposes of the designation of the United States of America:

I hereby declare that I believe I am the original inventor or an original joint inventor of a claimed invention in the application.

This declaration is directed to the international application of which it forms a part (if filing declaration with application).

This declaration is directed to international application No. PCT/US2022/044834 (if furnishing declaration pursuant to Rule 26ter).

I hereby declare that the above-identified international application was made or authorized to be made by me.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Name: Do-Gyoon KIM
Residence: Dublin, OH, USA
(city and either US state, if applicable, or country)

Mailing Address: c/o Ohio State Innovation Foundation
1524 North High Street
Columbus, OH 43201 United States

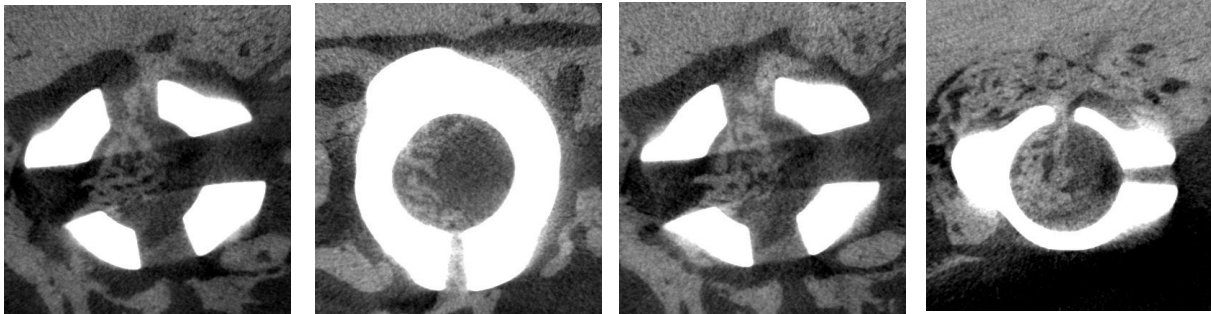
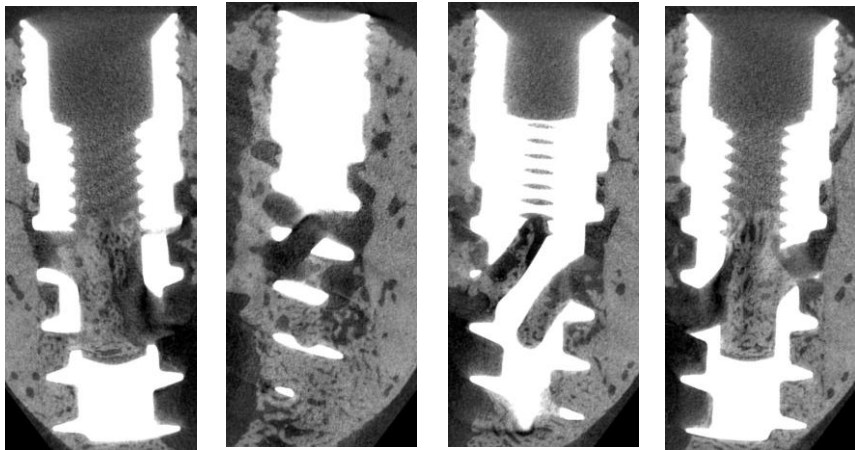
Inventor's Signature: [Signature] Date: 9/28/2022
(The signature must be that of the inventor, not that of the agent)

*PCT = Patent Cooperation Treaty: International filing system enabling patent protection across 150+ countries simultaneously

“Transforming 30 years of implant limitations into breakthrough patient solutions”

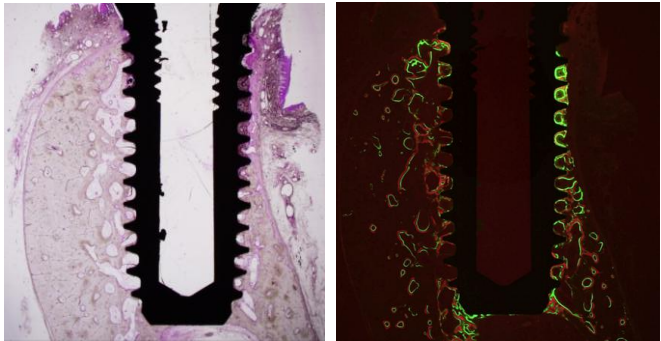
Redefining Implant Excellence: Safer, Faster, Simpler

3D Micro-Computed Tomography (10 micron) for bone ingrowth into implant at 6 weeks post-implantation



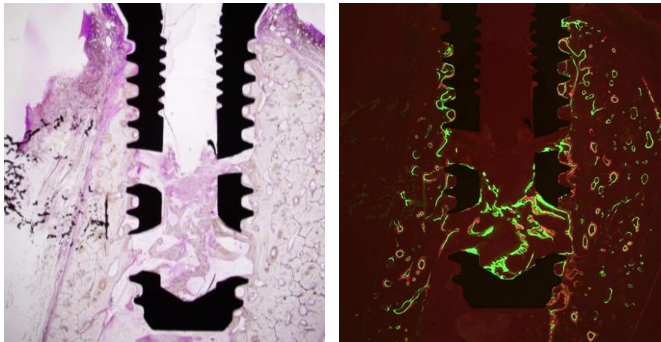
Canine model – 6wks post-implantation

Traditional implant

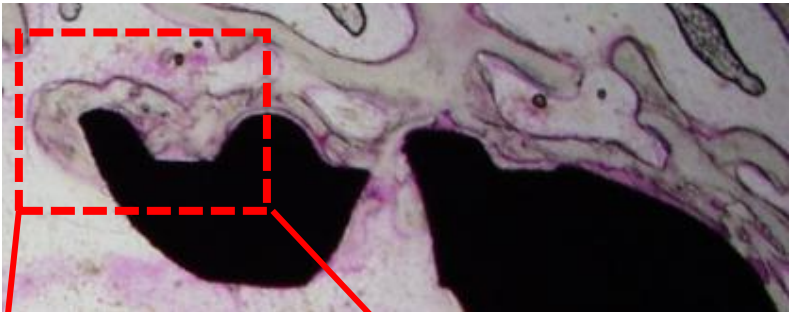


New bone ingrowth labels
(**alizarin (3wks)** and **calcein (6wks)**)

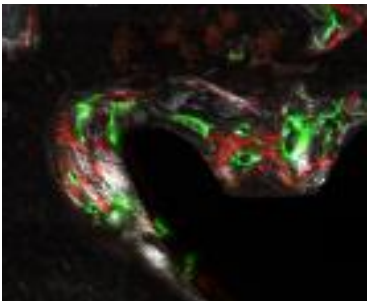
Hybrid Implant



Early bone ingrowth holds the implant to increase stability



Implant thread held by new bone



Alizarin & Calcein labels for new bone

≤ 140%

Improved Contact

Better bone-to-implant contact compared to traditional implants

\$3K+

Cost Savings

Eliminates bone grafting cost and 6-month waiting period

3D

Stronger Stability

Enhanced structural stability through 3-dimensional bone ingrowth

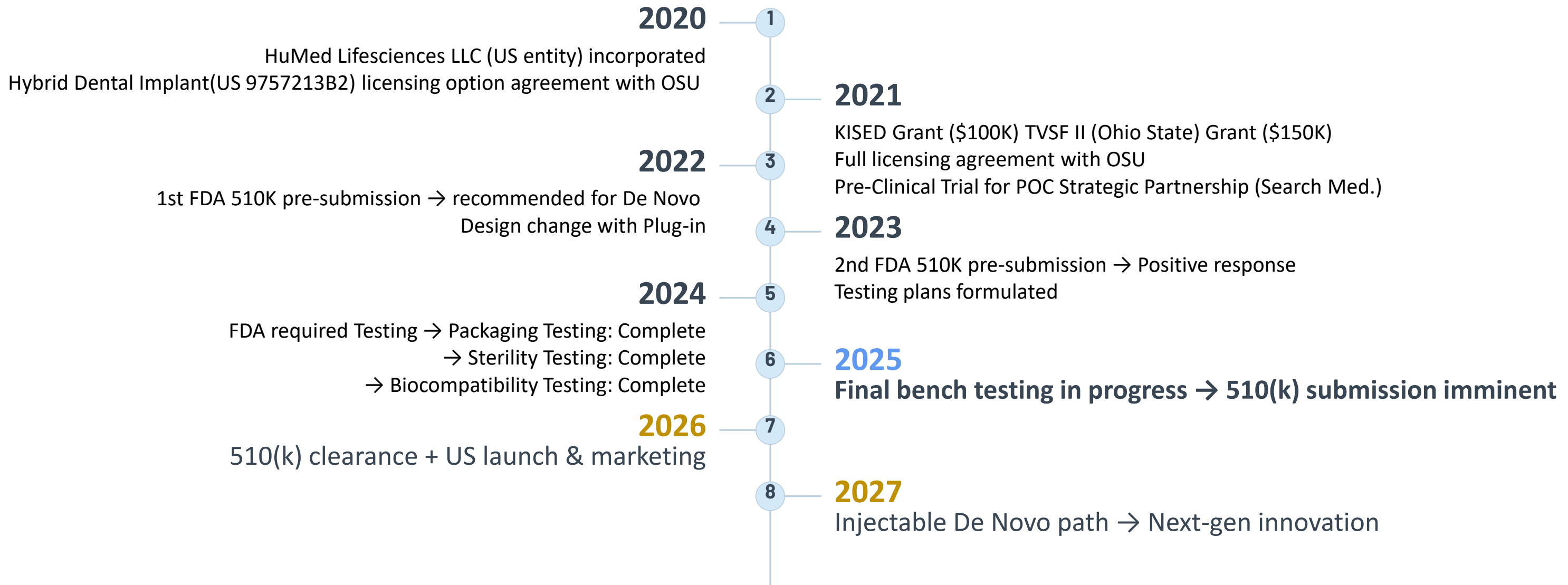
100%

Compatibility

Works with existing surgical instruments and procedures



Key Milestones on Our Journey



Innovation Excellence Powered by Market Mastery

Core Team



Dr. Do-Gyoon Kim, CTA, Founder

Ohio State University Professor
Inventor of Hybrid Implant (KiiS), 20+ years R&D



Hongsuk Kim, CEO

Global commercialization expert
ex-Samsung/J&J/P&G, Harvard A.M.

Advisory Board



Jinyong Oh, Strategic Advisor

J&J Medical Devices & P&G veteran
global scaling expert, MIT MBA



John Y. Hong

DDS, Serial Entrepreneur, Georgia Tech
Northwestern alum



Paul Fagan

Serial life science entrepreneur
ECM CEO, MIT MBA



Dr. Sreenivas Koka

Dean, Univ. of Mississippi Dentistry
ex-Mayo Clinic Chair

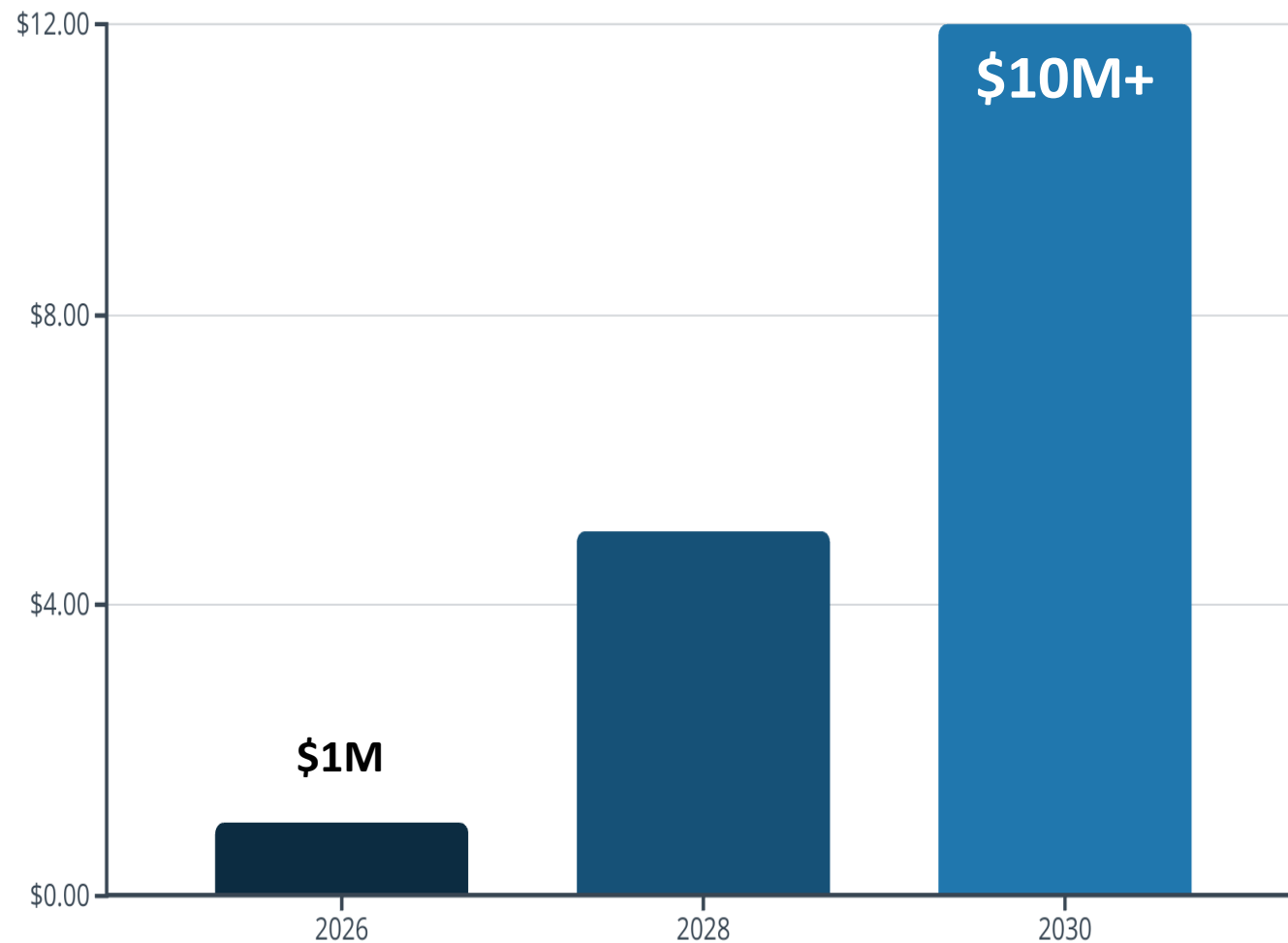


Tom Cole

VP, Dentsply Sirona & Inkbit (3D)
MIT MBA

Strategic Market Expansion: Path to \$10M Revenue

4-Year Path to \$10M+ Revenue



Go-To-Market Strategy (US)

Phase 1: Frictionless Adoption

Target: General Practice Dentists (27K)

Same workflow, same cost, revolutionary results

→ Higher margins & patient satisfaction



Phase 2: Gateway Enabler

Target: Non-implant General Dentists (108K)

Simplified procedures, enhanced outcomes

→ Start implants with confidence



Phase 3: Technical Supremacy & Premium Leadership

Target: Implant Specialists (8K)

Injectable innovation conquering complexity

→ Simplified procedures & clinical excellence



Ready to Revolutionize Implant Treatment Together?

Transforming patient outcomes through breakthrough technology

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